

Generative AI

How to Move from AI Experimentation to AI Transformation

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Illustration by Mandy Tsou



Summary. Many companies are investing heavily in AI but failing to translate isolated productivity gains into meaningful business results. The problem is a “micro-productivity trap,” where firms optimize tasks without rethinking workflows or value... [more](#)

Generative AI has sprinted from novelty to boardroom priority in record time. But despite the widespread adoption of this technology, not every company is realizing bottom-line improvement commensurate with its capabilities.

Our roles—serving as an AI provider to over one million businesses at OpenAI and guiding enterprise AI deployments at Bain—have given us unique insights into what approaches lead to transformative productivity lift and EBITDA growth and which ones fall flat. We’ve written this article for the many leaders whose firms are past the basic experimentation phase and are trying to deploy AI on a large scale but are seeing AI deployments get stuck or not yet deliver promised ROI.

Across financial services, retail, healthcare, technology, and other industries, we’ve observed that most firms that fail to capitalize on AI’s potential are stuck in what we call the “micro-productivity trap.” This means that they treat AI like a plug-and-play SaaS



investment with isolated use cases and scattershot pilots. In these cases, leaders use AI either to optimize existing offerings (“offering lock-in”) or to automate current processes without rethinking them (“process lock-in”), and often both. Individuals often realize productivity gains on key tasks, but those gains often stall at the firm level when the surrounding workflow still depends on tacit knowledge, manual handoffs, or legacy systems not built for AI. AI can accelerate a task, but unless companies address workflow bottlenecks, productivity gains may not translate into business value.

The firms that have been successful with AI transformation, on the other hand, take an organization-wide, future-focused perspective. They move away from “improve the task” to “reinvent the business.” Instead of optimizing for current offerings, they reexamine their core value proposition, reimagining where value lies and focusing on where they can win in the future. Instead of reinventing existing processes, they take an outcome-oriented approach, centering on what outcomes those processes serve and rebuilding their workflows starting



from the premise that we live in a world in which powerful AI tools exist.

This is where the real potential of AI lies: in a cross-functional reshaping of how your business grows its revenue, serves its customers, and competes in the marketplace. Those firms we've worked with that have embraced AI this way are seeing clear success metrics. Bain's clients have experienced 10–25% EBITDA gains that are continuing to increase as the programs scale. These results are delivering significant ROI on the underlying investment and effort.

We've observed that those companies that are successful in this effort have followed four common steps that have helped them to maintain the organization-wide, business reinvention mindset: 1) Narrowing possibilities and efforts strategically; 2) reimagining workflows across the organization, 3) engaging those closest to today's process to lead change, and 4) selecting the right measures of success. We'll show what each of these steps looked like for two of the firms we've worked with closely: home improvement retailer Lowe's (who partnered with OpenAI to launch the two offerings we highlight below) and a Fortune 1000 manufacturing company which we'll call FabricationCo (which Bain supported during the work we describe).



Step 1: Narrow Possibilities Strategically

AI's transformative potential is impacting all functions and roles, but the companies we've seen embrace AI most effectively have resisted the urge to spread it everywhere without real and specific outcomes in mind. Instead, they identify four or five critical domains and concentrate their transformation efforts there. Software development, customer support, knowledge worker efficiency, and marketing are consistently in the top four domains for AI use cases across Bain's client work and most frequently cited in decision-maker feedback, for example, but executives will need to find the right domains for their own industries.

To narrow the field to the right use cases for your firm, consider the organization as a whole and ask several questions:

- Which parts of the business will benefit most from AI today and tomorrow, assuming continued progress in capabilities?
- Where do we have the highest concentration of resources with the most repeatable work?
- What are the high-value, low-effort use cases (often where workers are bottlenecked today)?
- Which solutions should be purchased or built internally?



- Where do we have the skills and capabilities in-house and where do we need to partner?

For example, at FabricationCo, leaders assembled a high-performing team of frontline operators and managers from each division and convened a week-long cross-functional workshop focused on AI use case discovery. Over the course of the week, teams mapped the most important workflows end-to-end for both the customer and internal roles, surfacing clear operational pain points across teams such as sales, engineering, and manufacturing.

That effort surfaced 14 discrete AI use cases, representing tens of millions of dollars in aggregate potential value. This enabled the leadership team to focus on a small set of initiatives with clear potential for near-term results aligned to high customer impact and pain points shared across frontline teams. By focusing on a small subset of potential use cases, FabricationCo is now on track to realize about \$30M in additional profit.

In ramping up their AI efforts following the launch of ChatGPT, Lowe's focused on an area centered on the core value the company delivers: helping customers complete projects. Within their broader framework of "How customers shop with us, how we sell, how we work," they prioritized efforts based on "the maturity



of the technology, the size of the use cases, readiness for change and the dimensional risk, including brand risk as well as other business risks,” according to Chandhu Nair, SVP of stores, data, AI, and innovation. Given these factors, they landed on a vision of democratizing expertise across the organization and built two AI interfaces that launched in March of 2025: MyLow and MyLow Companion, which scaled expert home improvement knowledge to every online visitor and in-store associate, respectively.

Step 2: Reimagine Workflows Across the Organization

In our experience, it’s the process redesign—not the technology—that is the most challenging part of AI deployment for firms, and also creates most of the value. The full scope of the work across the company must be considered rather than having teams and departments approach process work in their own silos. Many seemingly department-specific work products are actually quite cross-functional. At FabricationCo, even seemingly straightforward activities like quoting customer jobs required coordination across sales, design, and operational teams, with meaningful process variation across areas of the business. This made these value streams prime candidates for reinvention rather than incremental automation.

As FabricationCo did, begin by understanding current workflows across the company for your chosen strategic priority and



determining how time is spent in that area, and by whom. This will help you begin to select the most fruitful opportunities from those you've identified, and ensure that the value is realized across the business. Ask: Which areas offer the highest value in terms of time, effort, and usage? Which processes are most ready in terms of repeatability, quality of supporting data, and technology? And where is there currently high variation across business units? Then reimagine these workflows with AI at the center.

From the broader set of opportunities they identified, FabricationCo used questions like these to shortlist two use cases to reimagine end-to-end, focusing on areas where manual effort, cycle time, and downstream costs were most acute. The first of these focused on rethinking the quoting process for customer bids. Previously, design engineers spent several hours producing initial designs as input for quotes for every bid, including those that were unlikely to convert, which was significant, at more than 50% for certain customer segments. In the transformed workflow, a rapid, 20-minute cost estimate is completed by non-designers for early-stage bids, allowing full engineered designs to be reserved for higher-probability opportunities. This shift reduced wasted effort while improving speed, focus, and overall bid economics, delivering a quote generation time roughly 15 times faster than the prior process. As a result, FabricationCo



significantly reduced total turn around time while also improving win rates.

Step 3: Engage Those Closest to Today's Process

Prescient leaders can describe what a new process will look like, its benefits, and even how it will feel different. But successful transformative solutions engage both top-down and bottom-up perspectives.

Redesigning processes requires close collaboration with leaders across seniority levels who are close to the work process but who are also committed to the AI transformation and able to envision a new way of doing the work. Individual contributors who are outstanding in their domain are also critical to deeply understanding the current process, especially those who can take a step back to reenvision how the work could get done more efficiently with AI. By involving front-line employees, leaders access insights from those closest to problem areas in the current approach who may have discovered individual productivity gains that can translate to broad workflow innovations.

Another way to engage employees is through prototyping. Encourage team members to join hackathons or micro-sprints. A culture of prototyping in all areas of the business (not just in the



tech team) can increase the velocity of iteration cycles and allow for bottom-up innovations to arise. The fruits of those sessions can be powerful: Offering team members even a simple wireframe of a new solution can energize potentially skeptical, anxious, or disengaged team members as they see what's possible and what's not, and where value in the organization will come from in the future.

FabricationCo redesigned workflows and supporting tools with a representative set of employees across regions from the start. Users were selected based on the strategic importance of their region and their willingness to be the first movers within the organization. Visible successes in piloted regions generated confidence in other regions ahead of the deployment to them. Even when the tool was piloted in a single region to test performance and gather real-time feedback, employees in other regions were also given demos and training on the tool and were invited to sit in on feedback sessions with employees in the pilot. This helped to enable fast scaling across other markets.

For Mylow Companion, Lowe's ran targeted store pilots with store associates to gather feedback from real-world usage. They began with one or two departments within each store (for example, plumbing or electrical), engaging store associates through in-app and on-floor feedback mechanisms to refine prompts, AI



guardrails, and UX, and then slowly expanding the testing across additional departments and stores. The development team utilized those insights to iterate on Mylow Companion before each expansion. Through that process, Mylow Companion has been fully deployed across over 1,700 Lowe's stores.

Step 4: Measure What Matters

When embarking on an AI initiative, it's critical to select the right metrics to track. "Efficiency" or "productivity" are too vague; instead, those benefits should be tied to key business outcomes, with metrics that allow for comparisons with non-AI methods.

For example, FabricationCo evaluated its redesigned workflows primarily through the lens of customer responsiveness and market differentiation. The overarching objective was to leverage AI to deliver faster quotes, ultimately improving win rates. To assess progress against this goal, the company focused on several key metrics: win rates for AI-generated quotes versus non-AI quotes, quote turnaround times, margins on downstream material and factory costs, and the volume and accuracy of priced bids. Across the two prioritized workflows, the company saw meaningful improvement on each of these dimensions—most notably, a 10-percentage-point increase in win rate within the first three months of deployment—reinforcing the case for scaling these changes across the enterprise.



Additionally, AI systems require new forms of sustained measurement. Because these systems, like humans, can produce different results from the same request, you need to evaluate their behavior continuously. To do this, an evaluation suite (“eval”) is required; evals measure your system’s output against an acceptable range of desired behaviors. The process for building them requires not only a clear definition of those expected behaviors but also identifying appropriate tolerance levels, auditing your processes to ensure they comply, and improving them when they do not.

To evaluate their system, Lowe’s engaged subject matter experts to create a set of prompts and expert-validated responses for how Mylow Companion should reply to typical questions posed by an associate. They evaluated how various models performed against the validated responses and fine-tuned their prompting to improve system performance. In addition, Lowe’s also collected user feedback and had its associates regularly audit both the outputs shown to customers and the system’s intermediate steps to identify opportunities for improvement and provide confidence on whether the system was meeting expectations.

To measure outcomes, Lowe’s chose to track drivers of business results, such as basket size, conversation rate, and incremental sales, as well as key drivers of success such as associate



knowledge, customer satisfaction, and ease of check-out. When customers engage with Mylow during their online visits, the conversion rate more than doubles and Lowe's is seeing customer satisfaction scores increase 200 basis points when associates use Mylow Companion to help customers shopping in the aisle.

The Boardroom Imperative

C-suite leadership is critical for the success of AI transformation. In companies that lag behind in AI adoption, we've observed that leaders recognize AI as an important opportunity, but they relegate responsibility for it to technology groups or divisions. Typically, this delegation comes without specific goals or metrics, just a vague "improve productivity" or "focus on high potential sources of value." From there, initiatives often fail.

Instead, AI transformation must be prioritized at the very top levels of the company, with all executives and functions participating, because leaders who have an organization-wide view are critical to transformative deployments. They must set ambitious, top-down, business-owned targets and then keep their teams accountable.

Going from lagging to leading requires that companies avoid siloed initiatives and enable instead a more wholesale



transformation of their business. To that end, these steps can help leaders and organizations out of the limited impact of the micro-productivity trap.

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